19 Given that $x^2 - 6x + 1 = (x - a)^2 - b$ for all values of x,

(i) find the value of a and the value of b.

(i) $x^2 - 6x + 1$

$= (x - 3)^2 - 9 + 1$

$= (x - 3)^2 - 8$

1 mark
Half of −6 is −3.

1 mark
a = 3, b = 8.

$a =$ 3

$b =$ 8

(2)

(ii) Hence write down the coordinates of the turning point on the graph of $y = x^2 - 6x + 1$

(ii) Minimum when x = 3: y = −8.

1 mark
Turning point (3, −8).

(3, −8 ,)

(1)

(Total for Question 19 is 3 marks)